Air Quality Companion – Project Report

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1. Introduction

Air pollution is a major environmental and health concern, especially in rapidly urbanizing regions. Poor air quality can lead to serious health issues such as respiratory diseases, cardiovascular problems, and reduced quality of life.

The **Air Quality Companion** project is designed to address this issue by providing users with real-time air quality analysis and personalized health recommendations. By leveraging historical air quality data from Indian cities, the system helps users make informed decisions about their daily activities.

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2. Objectives

The main objectives of this project are:

* To analyze air quality using real-world data
* To calculate Air Quality Index (AQI) based on pollutant levels
* To provide personalized health recommendations
* To suggest suitable daily activities based on air conditions
* To compare air quality across different cities
* To create a user-friendly, conversational interface

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3. Scope of the Project

The project focuses on:

* Processing historical air quality data (2015–2020)
* Estimating AQI using PM2.5 and PM10 values
* Providing health and activity suggestions
* Enabling quick AQI lookup for integration with other systems

Limitations:

* Does not use real-time API data (only historical dataset)
* No machine learning model implemented (rule-based system)

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4. System Architecture

The system follows a simple architecture:

1. **Input Layer**

   * User inputs PM2.5 and PM10 values

2. **Processing Layer**

   * AQI calculation using standard formulas

* Matching with historical data
* Generating recommendations

3. **Output Layer**

* AQI value
* Health advice
* Activity suggestions

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5. Technologies Used

Technology	Purpose
Python 3.6+	Core programming language
Pandas	Data manipulation and analysis
NumPy	Numerical computations
CSV Files	Dataset storage and processing
Kaggle Dataset	Source of air quality data

7. Methodology

Step 1: Data Collection
Step 2: Data Preprocessing

* Load dataset using Pandas
* Handle missing values

* Filter relevant columns (PM2.5, PM10)

Step 3: AQI Calculation

* Use standard AQI formulas
* Convert pollutant concentrations into AQI values

Step 4: Pattern Matching

* Compare current input with historical data
* Identify similar pollution patterns

Step 5: Recommendation Generation

* Based on AQI levels:

  * Good → Outdoor activities encouraged
  * Moderate → Limited outdoor exposure
  * Poor → Stay indoors
  * Severe → Avoid outdoor activities

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8. Features

* Real data analysis using Indian city dataset
* AQI estimation based on pollutant levels
* Intelligent matching with historical patterns
* Personalized health recommendations

* Activity suggestions based on air quality

* City comparison functionality

* Quick AQI lookup function

* Conversational interface

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9. Implementation

Installation

```bash
pip install pandas numpy
```

Run Application

```bash
python air_quality_companion.py
```

Quick Function Example

```python
from air_quality_companion import quick_air_lookup

result = quick_air_lookup(35, 65)
print(result['aqi'])
print(result['suggestion'])
```

```

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## **10. Testing**

### **Test Case 1: Basic Functionality**

* Input: PM2.5 = 25, PM10 = 50
* Expected: AQI + suggestions displayed

### **Test Case 2: Different Scenarios**

| Scenario | PM2.5 | PM10 | Expected Result |
| --------- | ----- | ---- | --------------- |
| Clean | 15 | 30 | Good AQI |
| Moderate | 45 | 80 | Moderate AQI |
| Poor | 120 | 200 | Health warning |
| Dangerous | 300 | 400 | Severe warning |

### **Test Case 3: Quick Lookup**

* Function returns AQI and recommendation

### **Test Case 4: Data Validation**

* Ensure dataset loads correctly
* Verify AQI calculations
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11. Results

Example Outputs:

Good Air Quality

* AQI: 42
* Suggestion: Outdoor activities recommended

Poor Air Quality

* AQI: 185
* Suggestion: Stay indoors, use air filters

The system successfully provides:

* Accurate AQI estimation
* Clear health guidance
* Practical activity suggestions

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12. Advantages

* Simple and user-friendly
* Uses real-world data
* Fast and efficient

* Easily extendable

* Helpful for daily decision-making

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13. Limitations

* No real-time data integration

* No machine learning model

* Limited pollutants (only PM2.5, PM10)

* Static recommendation logic

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14. Future Enhancements

* Integration with real-time AQI APIs

* Machine learning-based prediction models

* Mobile or web application interface

* Personalized recommendations based on user health data

* Forecasting future air quality trends

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15. Conclusion

The Air Quality Companion project successfully demonstrates how data analysis can be used to solve real-world problems. It provides meaningful insights into air quality and helps users take preventive actions to protect their health.

Although currently rule-based, the system has strong potential to evolve into a full AI-driven application with predictive capabilities and real-time data integration.

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16. References

* Kaggle: Air Quality Data in India (2015–2020)

* Standard AQI Guidelines (India)

* Python Documentation

* Pandas & NumPy Documentation
